TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Surface Tension of Mercury and the Angle of Contact by Quincke’s Method

A large mercury drop flattens under its own weight until surface tension and hydrostatic pressure balance — its profile height gives $T = \frac{1}{2}\rho g h^2$, and its edge gives the contact angle.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 9 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find the surface tension of mercury and its angle of contact with glass from the shape of a large sessile drop. A big drop of mercury on a clean glass plate spreads until gravity (which flattens it) and surface tension (which holds its edge) balance. The top of the drop is nearly flat; at the level of the drop’s greatest horizontal cross-section the surface is vertical. Quincke showed that if h is the vertical distance from the flat top to that widest level, then

$$T = \tfrac{1}{2}\,\rho\,g\,h^2$$

The **angle of contact** θ is read from the drop’s profile at the plate: the angle, measured *through the liquid*, between the plate and the tangent to the drop’s surface at the edge. For mercury on glass θ is obtuse (about 140°) — mercury does not wet glass (Physics-I Topic 3), which is exactly why it forms a proud drop instead of a climbing film, and why the capillary method of Experiment 6 cannot be used for it in the same way (mercury is *depressed*, not raised, in a capillary).

2. Apparatus and procedure

- **Apparatus:** a pool/drop of clean mercury on a levelled glass plate, travelling microscope, spirit level, glass-cleaning reagents; the plate and mercury must be scrupulously clean (oxidised or dusty mercury gives a ragged profile).
- Form a large drop (several cm across — large enough that the top is effectively flat) on the plate; let it settle; level the plate.
- With the travelling microscope, read the vertical position of the flat top of the drop, then of the level of greatest width (the vertical tangent level); h is the difference.

- Measure the profile near the edge (microscope readings of the tangent point and the plate surface) and determine θ from the tangent at the contact line; alternatively read θ with the microscope's rotating cross-wire aligned to the edge tangent.
- Compute $T = \frac{1}{2}\rho g h^2$ with $\rho_{Hg} = 13{,}600$ kg m^{-3}.

3. Worked calculation (model data)

RESULT

Microscope readings: top of drop 3.412 cm; widest level 3.148 cm → h = 0.264 cm = 2.64×10^{-3} m.
$T = \frac{1}{2} \times 13{,}600 \times 9.8 \times (2.64 \times 10^{-3})^2 = 66{,}640 \times 6.97 \times 10^{-6} =$ **0.465 N m^{-1}** — mercury's accepted surface tension (about 0.465–0.472 N m^{-1} at room temperature), some six times water's. Angle of contact from the edge tangent: θ = **140°** (through the mercury), the textbook value for clean mercury on clean glass. The two results tell one story: a high-T liquid that refuses to wet the plate stands in a tall, blunt-edged drop.

4. Precautions

- Mercury hygiene: work over a tray, never with bare hands in the vapour path for long, collect every spill droplet, and never heat mercury — its vapour is toxic. Spilt mercury in cracks is removed with a sulphur dusting or a mercury spill kit, per laboratory rules.
- The drop must be large: in a small drop the top never flattens and the formula does not apply (surface tension then wins everywhere and the drop is a spherical cap).
- Cleanliness again decides: oxide skin on mercury pins the edge and distorts both h and θ; filter or use fresh, bright mercury.
- Readings of the two levels must be taken with the microscope moving in one direction only (backlash), and the plate must stay levelled throughout.
- Allow the drop to settle after forming — a still-spreading drop has a moving contact line and a false profile.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does the formula involve h² rather than h, and what balance of forces does it encode?
2. A mercury drop gives h = 2.50 mm. Compute T.

3. Define the angle of contact, and state why mercury's on glass is obtuse while water's is (nearly) zero.
4. Why must the drop be large? What shape does a very small mercury drop take, and why?
5. What would a thin oxide skin on the mercury do to the measured h and θ?
6. Why can the capillary-rise formula of Experiment 6 not simply be reused for mercury with a "negative rise"?

Answer key

1. At the drop's widest level the hydrostatic pressure inside (ρgh, growing linearly with depth below the flat top) is balanced by the Laplace pressure of the curved surface there, whose radius of curvature is set by the drop's own height. Balancing a pressure $\propto h$ against a curvature pressure $\propto T/h$ gives $h^2 \propto T/(\rho g)$ — Quincke's full analysis of the profile fixes the constant: $T = \tfrac{1}{2}\rho g h^2$. The h^2 thus encodes depth and shape in a single measured length.
2. $T = \tfrac{1}{2} \times 13{,}600 \times 9.8 \times (2.50 \times 10^{-3})^2 = 66{,}640 \times 6.25 \times 10^{-6} = 0.417$ N m^{-1} — below the clean-mercury value; in the laboratory this is the classic signature of slightly contaminated mercury (Q5).
3. The angle of contact is the angle between the solid surface and the tangent to the liquid surface at the contact line, measured through the liquid. It is set by the balance of adhesive (liquid–solid) and cohesive (liquid–liquid) forces: water adheres strongly to glass and spreads ($\theta \approx 0°$, acute); mercury's cohesion far exceeds its adhesion to glass, so it beads up with $\theta \approx 140°$, obtuse.
4. Only a large drop develops the flat top and the vertical-tangent widest level that the derivation assumes — gravity must be strong enough, over the drop's size, to flatten it against surface tension. A very small drop is dominated by surface tension everywhere and takes a spherical-cap shape, for which h measures the cap's geometry, not the Quincke balance.
5. The skin stiffens and pins the surface: the drop cannot relax into its true equilibrium profile, the edge tangent is ragged (θ unreliable, usually read too large or scattered), and h is distorted — the remedy is fresh, bright, filtered mercury and a freshly cleaned plate.
6. Because mercury's meniscus in glass is convex with $\theta = 140°$, the capillary formula $T = \rho g r h/(2\cos\theta)$ breaks down in practice: the "depression" is small, the meniscus shape hard to read, and the $\cos\theta$ factor with an ill-defined angle on an oxidising surface makes the method unreliable. Quincke's method measures mercury on its own terms, through its drop profile.
